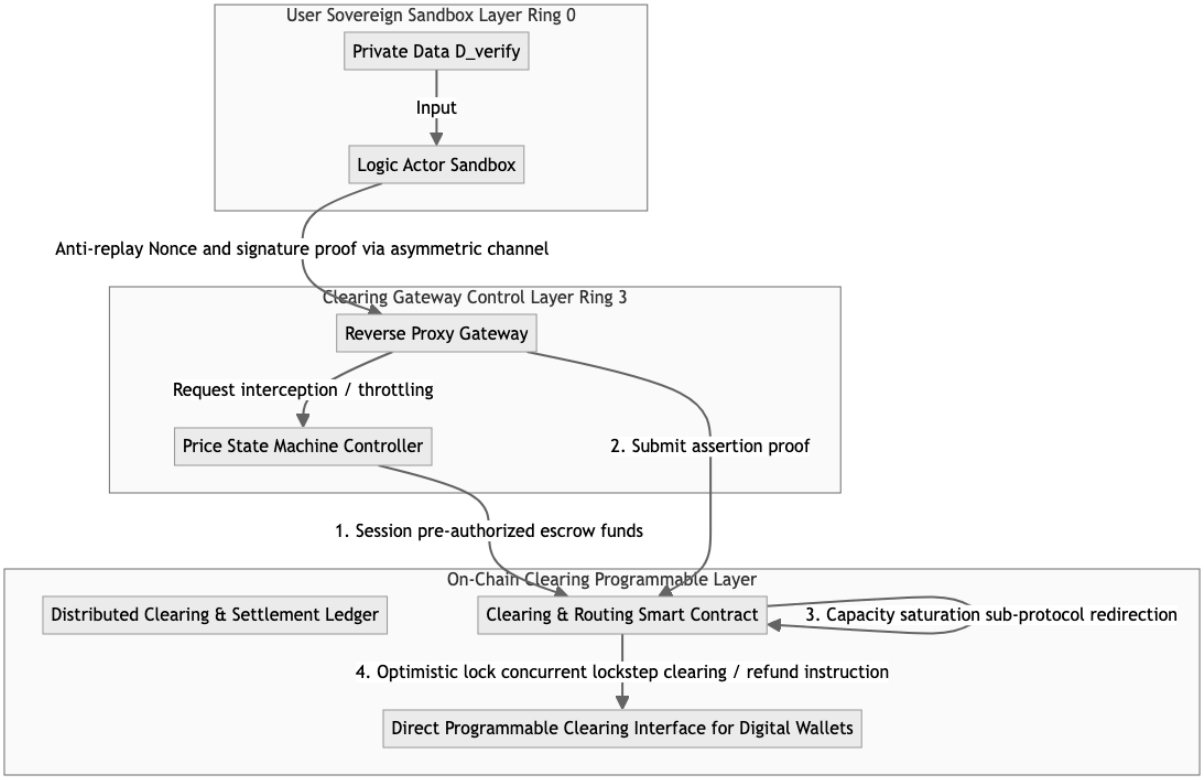

User Sovereign Sandbox Layer Ring 0
Private Data D_verify
Input
Logic Actor Sandbox
Anti-replay Nonce and signature proof via asymmetric channel
Clearing Gateway Control Layer Ring 3
Reverse Proxy Gateway
Request interception / throttling
Price State Machine Controller
2. Submit assertion proof
1. Session pre-authorized escrow funds
On-Chain Clearing Programmable Layer
Distributed Clearing & Settlement Ledger
Clearing & Routing Smart Contract
3. Capacity saturation sub-protocol redirection
4. Optimistic lock concurrent lockstep clearing / refund instruction
Direct Programmable Clearing Interface for Digital Wallets

图1

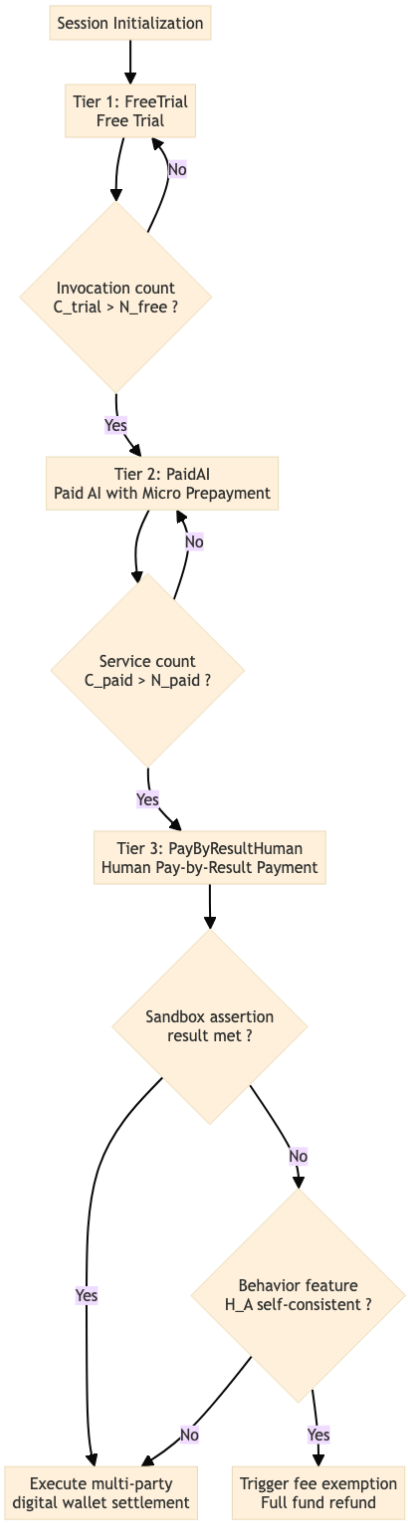

Session Initialization
Tier 1: FreeTrial
Free Trial
No
Invocation count
C_trial > N_free ?
Yes
Tier 2: PaidAI
Paid AI with Micro Prepayment
No
Service count
C_paid > N_paid ?
Yes
Tier 3: PayByResultHuman
Human Pay-by-Result Payment
Sandbox assertion
result met ?
No
Yes
Behavior feature
H_A self-consistent ?
No
Yes
Execute multi-party
digital wallet settlement
Trigger fee exemption
Full fund refund

图2

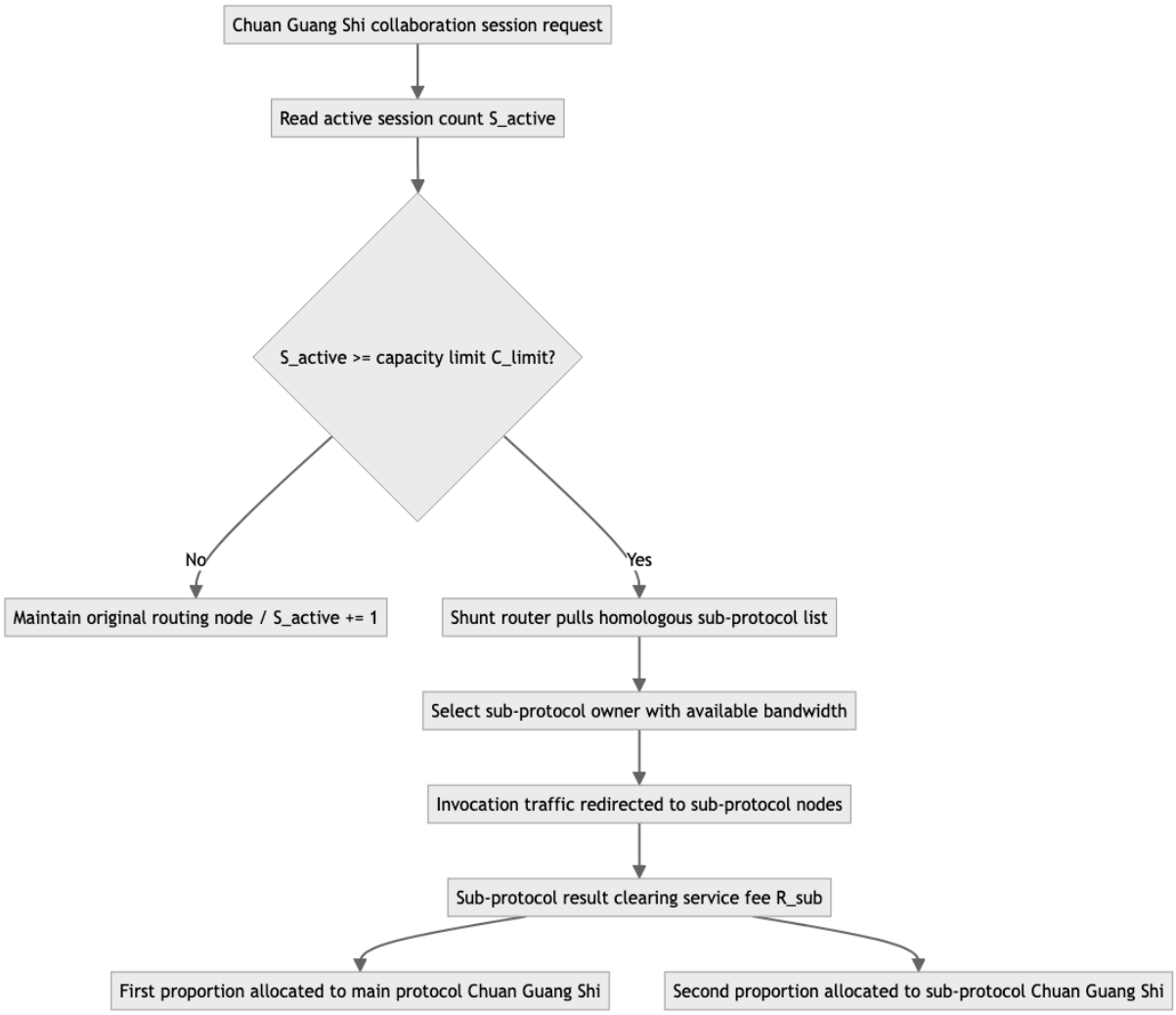

Chuan Guang Shi collaboration session request
Read active session count S_active
S_active >= capacity limit C_limit?
No
Yes
Maintain original routing node / S_active += 1
Shunt router pulls homologous sub-protocol list
Select sub-protocol owner with available bandwidth
Invocation traffic redirected to sub-protocol nodes
Sub-protocol result clearing service fee R_sub
First proportion allocated to main protocol Chuan Guang Shi
Second proportion allocated to sub-protocol Chuan Guang Shi

图3